Weekly Report

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Review

- Started looking at GEM and trigger multiplicities for cosmic and beam data
  - Created new histograms
  - GEM HV settings were not consistent in runs 4026 and 2792
  - Run 4026 was elastic + moller

This week

- Pushed changes to darklight-cooker
  - GEMTrack now includes “out of time” hits and tracks
  - TrigScint now includes “out of time” hit multiplicity histogram
- Trying to find good data runs
- Working on documentation

Elastic Line Data

- Dark current background
  - Beam veto fully implemented on Dec 16 (E-log: 66)
  - Data before Dec 16 is contaminated with dark current background?
  - Veto timing fixed on Dec 23 (E-log: 75) – dark current “only” runs included beam?

Elastic Line Data

- Good data: Dec 16 2025 – Jan 21 2026

Elastic Line Data

- Spectrometer background
  - Too much background with both dipoles on? (E-log: 69 Dec 17)
  - We switched to elastic line magnet setting on one side at a time

To-Do List

1. Detector multiplicity study
2. Analysis code development
3. Event display

1. Detector Multiplicity Study

a) Understand event and hit timing
   - Look at timing of trigger hits per event, in-time versus out-of-time

b) Understand GEM cluster identification
   - Look at clusters which do not meet quality threshold

c) Improve histograms/plots
   - Look at detector multiplicities per events with trigger

d) Compare cosmic vs elastic-line data

Bonus: Compare beam data with different GEM HV settings

2. Analysis Code Development

a) Trigger hit selection
- Out-of-time hit cuts?

b) GEM cluster selection
- Adjust quality thresholds, charge calculation?
- Correlate X and Y signals based on cluster charge

c) Detector track selection
- Trigger hit non-dispersive position
  - Look at trigger hits vs track position X and Y
  - Fix calibration issue
- Implement timing information?